Corpus Verification & Validation

- Progress report to Supervisor

2,400

corpus-level measurements

5

defect classes injected

1.000

AUC, document-level detection

Where we started

Assessment of the existing draft

1 Three half-papers in one file

The divergence metric appears twice under two names (E and M) with identical worked examples, tables and figures duplicated.

2 Claims without evidence

Numbers (0.84 vs 1.84) were shown to be plausible — never shown to be correct. No ground truth existed.

3 Announced but undelivered

Two of three new corpora had no results; the ten-step alignment pipeline had no experiments at all.

The core gap: illustration is not evaluation

What the draft did

- Compute divergence between Brown and LOB → small number
- Compute it between Brown and the Bible → larger number
- Conclude the measure “behaves sensibly”
- **But: no defect was ever introduced, so nothing was detected**

What we did instead

- Start from a clean corpus
- Inject a known defect at a known rate
- The correct answer now exists by construction
- **Measure detection: dose–response, effect size, ROC-AUC, vs a baseline**

What the paper now contributes

1

Framework made testable

The verification/validation split is recast as a prediction about which instrument detects which defect, not just a taxonomy.

Prediction confirmed

2

Metric corrected

The measure called “relative entropy” was neither KL divergence nor a metric. Replaced with smoothed Jensen–Shannon divergence.

√JSD: 60/60 metric checks

3

Ground truth built

First controlled defect-injection benchmark for corpus V&V: 5 defect classes × 8 rates × 60 runs.

2,400 measurements

Correcting the metric

The quantity in use was mis-identified in the literature

Property	Draft measure	Corrected (JSD)
Is it KL divergence?	No: unweighted log-ratios	Yes: probability-weighted
Symmetric?	Yes (by absolute value)	Yes (by construction)
Bounded?	No	Yes: 0 to 1 bit
A true metric?	Claimed, never proven	Yes: JSD, 60/60 checks
Zero-frequency problem	Open limitation	Dissolved by smoothing

Consequence: “distance metric” becomes a defensible claim.

Building the ground truth

Controlled defect injection, the correct answer is known by construction

1

Clean corpus

9 public-domain texts
(English, Bible, Arabic)

→

2

Split

Trusted REFERENCE half
vs TEST half

→

3

Inject

Known defect at
known rate r

→

4

Measure

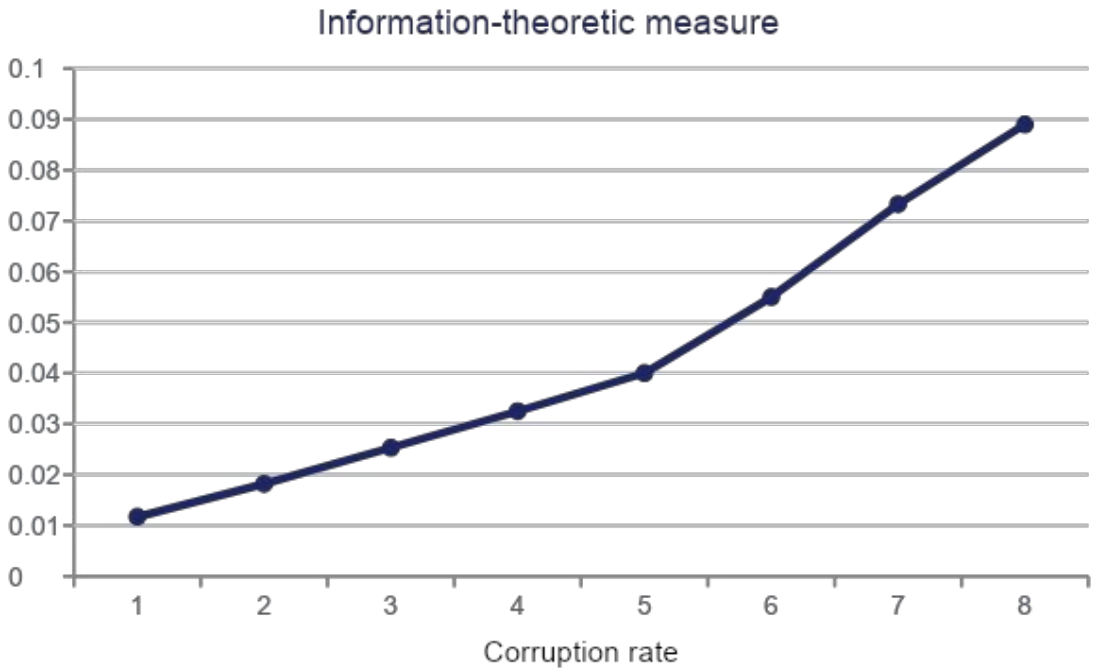
Divergence vs baseline
n = 60 per cell

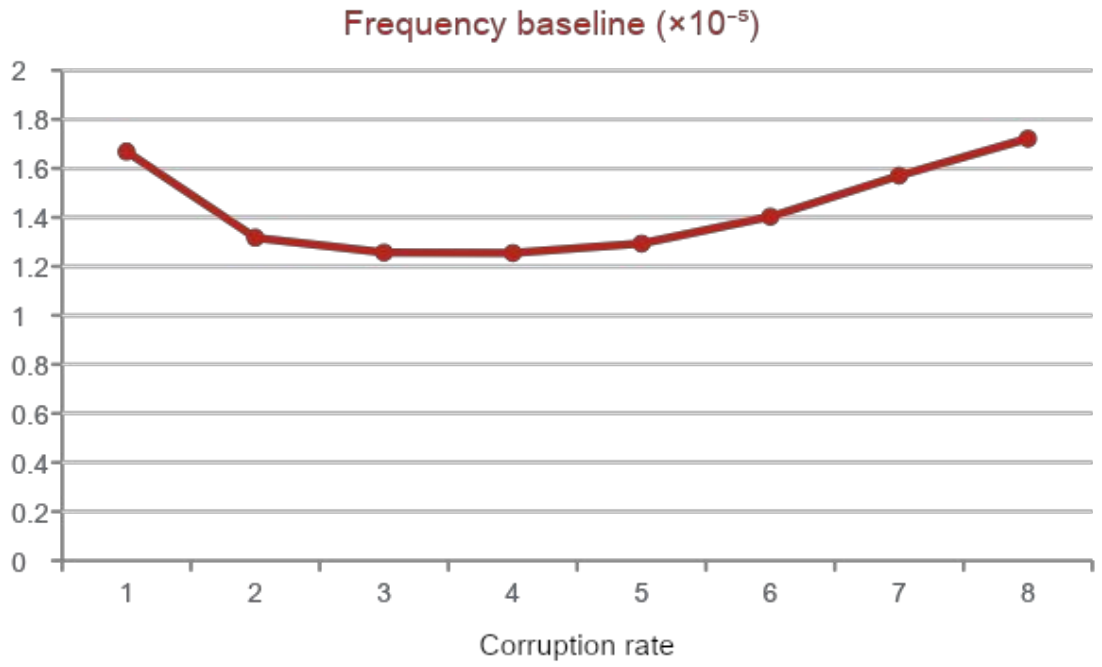
Five defect classes

duplicate · truncate · boilerplate | foreign text · encoding corruption

Headline result: the standard baseline fails

Encoding corruption, the frequency measure moves the wrong way

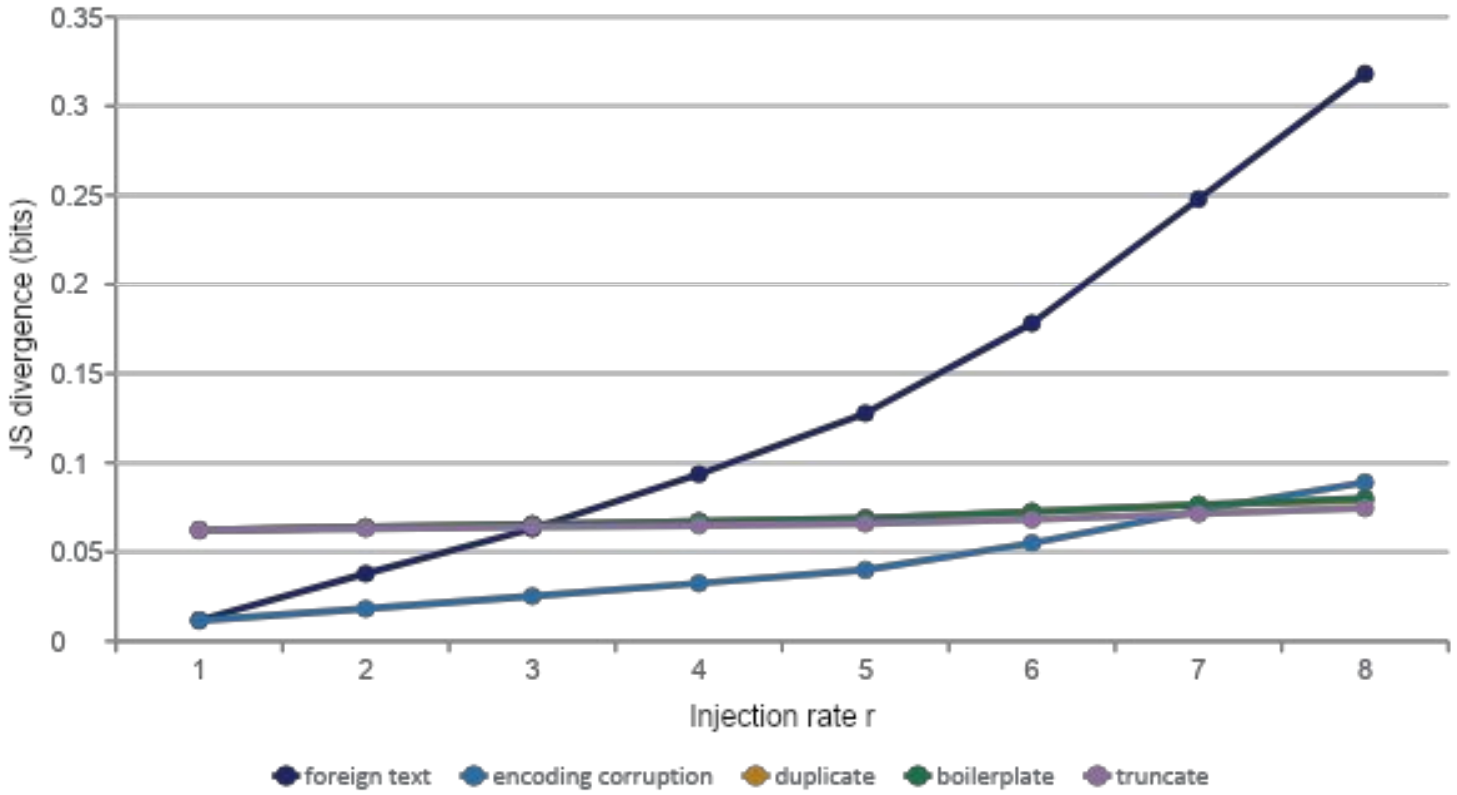




Baseline effect size d = −0.80, p = 1.000. It reports a more corrupted corpus as ***more similar*** to its clean reference. The corrected measure detects the same defect at $p \approx 3\times10^{-12}$.

The measure responds to every defect

Monotonic in injection rate, 7/7 rate-steps, all five defect classes



Reading the chart

- Content defects (top two lines) rise steeply and separate cleanly from the r = 0 null.
- Structural defects (bottom three) rise only shallowly, they are barely visible at corpus level.
- **That flatness is not a failure of the metric. It is exactly what the framework predicts.**

The framework makes a correct prediction

Structural defects need a different instrument and the theory said so in advance

Corpus-level instrument

Structural defects at r = 0.05

all p > 0.11

Not detected. Duplication, truncation and boilerplate are invisible to aggregate divergence.

Document-level instrument

Same defects, same corpora

AUC = 1.000

Perfect detection, zero variance, across 360 runs at every injection rate tested.

This converts the verification/validation distinction from a descriptive taxonomy into an operational one: it tells a practitioner which instrument to reach for.

Three bugs caught, and why we report them

Each produced a plausible result that was wrong

Segmenter returned 1 document

The Arabic text has no blank lines, so it loaded as a single 706k-character document. Foreign-text injection was inserting an entire book.

Fake signal: 0.96 True: 0.088

Tokenizer destroyed Arabic

Python's \w does not match Unicode combining marks. Diacritics split every word into letters: 285,735 "tokens", only 207 types.

After fix: 77,429 tokens / 21,117 types

Ground-truth labels were wrong

Only the overwritten document was labelled a duplicate, not the one it was copied from, penalising a correct detector.

AUC 0.947 → 1.000

All three are verification failures in our own pipeline: invisible to aggregate statistics

Status and next steps

Completed

- Every number audited against source data
- Key citations verified, not recalled
- Full pipeline reproducible

Next

- Run the experiment on different publicly available corporas and compare results
- Swap in the real BACC / CCA corpora, and see what changes
- Fix overleaf issue and start writing the first draft of the paper

The paper will no longer asks to be believed. It will show what it detects, how well, and where it fails.